BO_correlations

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```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.4.4     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.0
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(corrplot) #Correlation Matrix
```

```
## corrplot 0.92 loaded
```

```r
library(reshape2)
```

```
##
## Attaching package: 'reshape2'
##
## The following object is masked from 'package:tidyr':
##
##     smiths
```

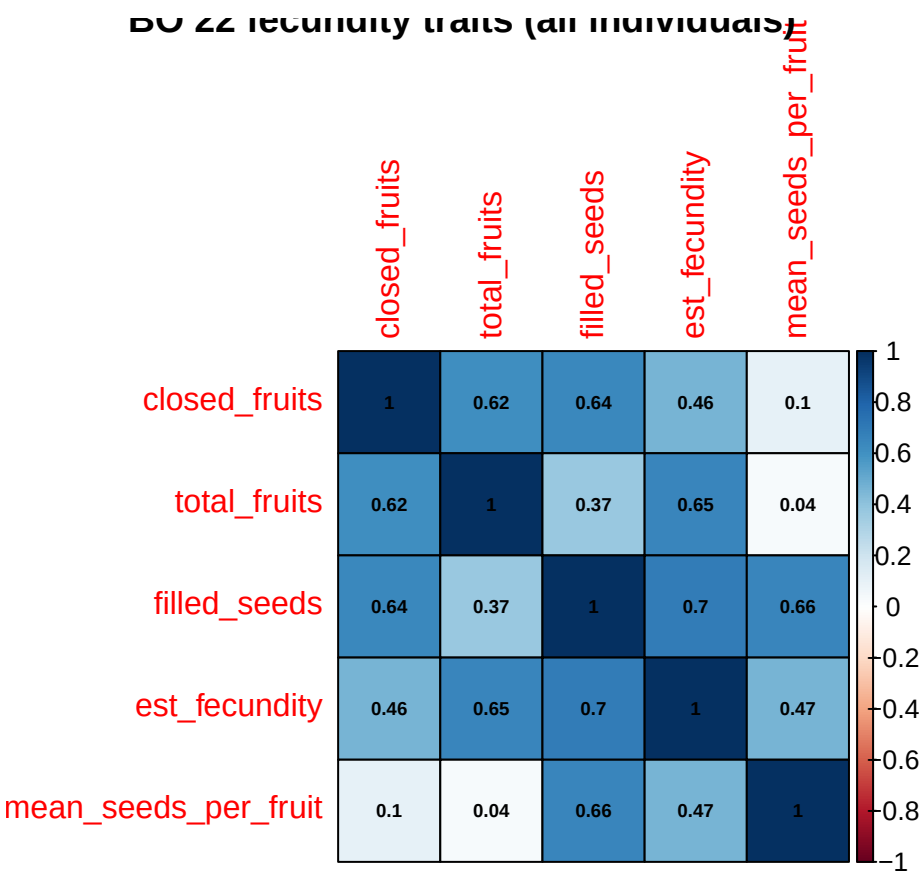
```r
# Load and preprocess data
# Read the CSV file and create 'sample_ID' column
BO_fit <- read_csv(here::here("data_sheets", "compiled_sheets", "BO_22_23_fit.csv")) %>%
  rename_at(vars(49), ~"surv_to_fruitpod") %>%
  mutate(
    # Construct sample_ID conditionally
    sample_ID = case_when(
      is.na(SegPos) ~ paste(Year, Recipient, Gen, str_pad(Transect, width = 3, pad = "0"), Plant_ID, sep
      TRUE ~ paste(Year, Recipient, Gen, str_pad(Transect, width = 3, pad = "0"), SegPos, Plant_ID, sep
    ),
    # Replace leading underscores in sample_ID with hyphens
    sample_ID = str_replace(sample_ID, "^([^_]*)_", "\\1-")
  )
```

```
## Rows: 596 Columns: 67
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (9): Gen, Donor, Recipient, sample_ID_SEG, sample_ID, SegPos, F_Num_03...
## dbl  (48): Year, Transect, Sequence, Plant_ID, days_sow2flower, days_plant2f...
```

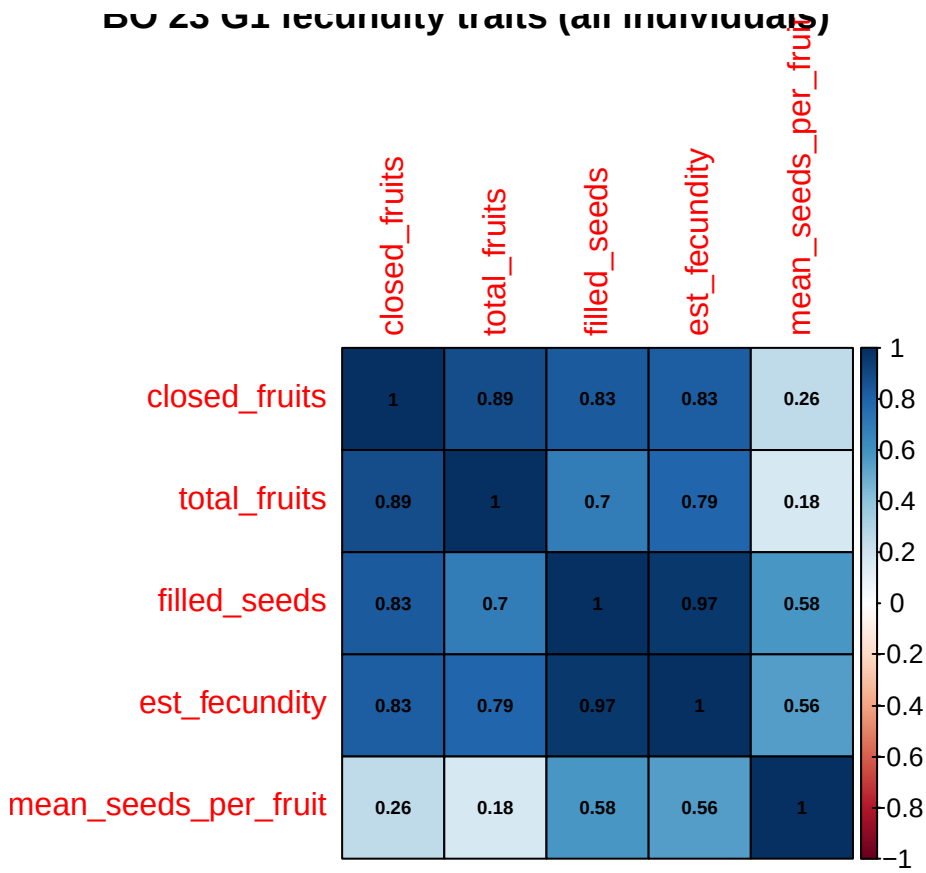
```
## lgl   (3): F_plant, Rep_FitP, any_FitP
## date  (7): Sow_Date, Plant_Date, FFD, LFD, photo_date, photo_date_SEG, Germ_...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
BO_fit_22 <- BO_fit %>%
  filter(Year == "2022")

BO_fit_23_G1 <- BO_fit %>%
  filter(Year == "2023", Gen == "G1")

BO_fit_23_G2 <- BO_fit %>%
  filter(Year == "2023", Gen == "G2")
```

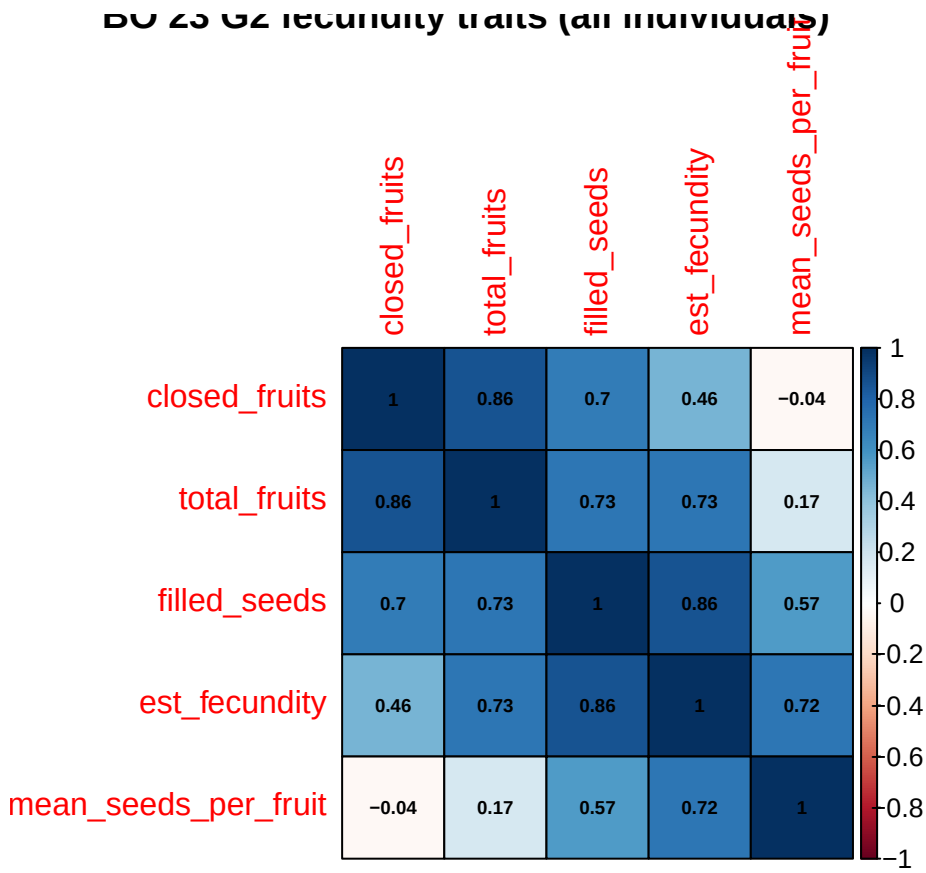
```
##BO 22: all individuals
corrplot(cor(BO_fit_22 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundity, mean_seeds_p
```



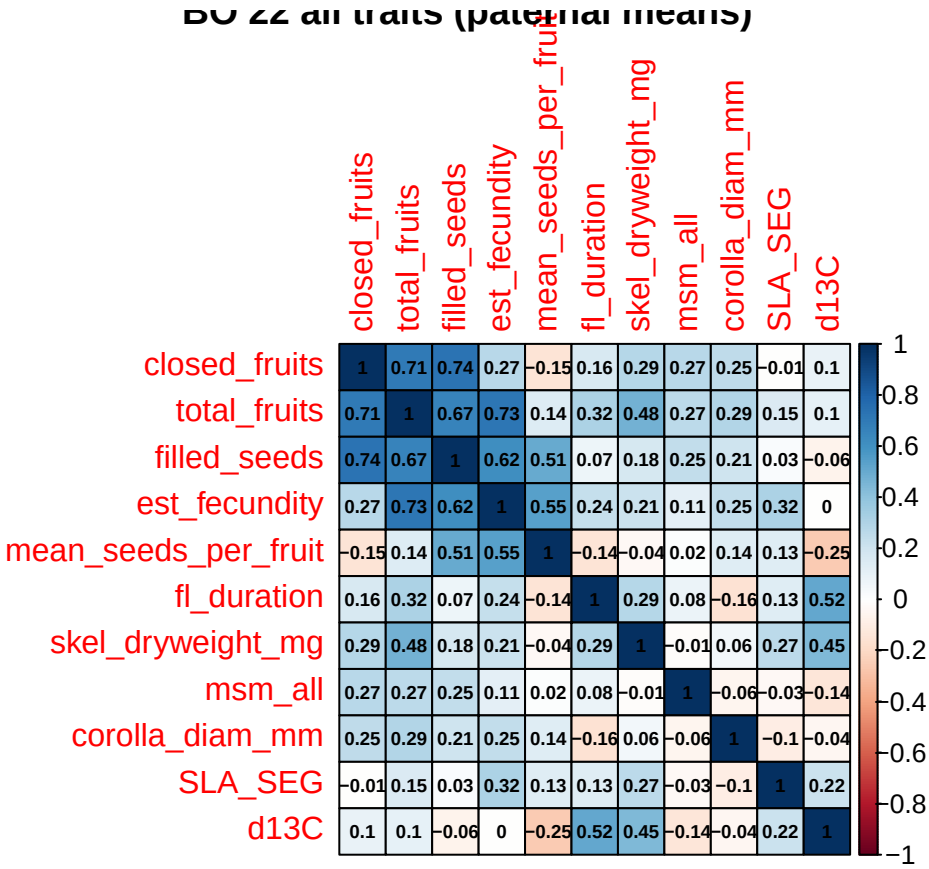

```
##BO 23 G1: all individuals
corrplot(cor(BO_fit_23_G1 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundity, mean_seed
```



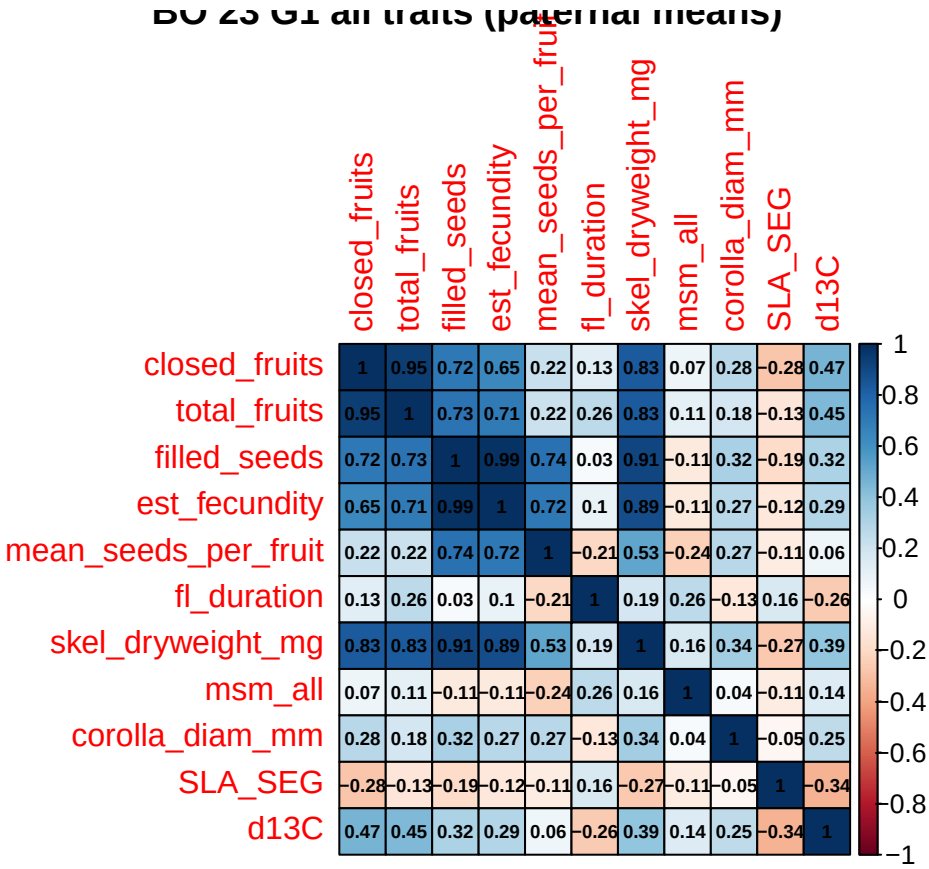

```
##BO 23 G2: all individuals
corrplot(cor(BO_fit_23_G2 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundity, mean_seed
```



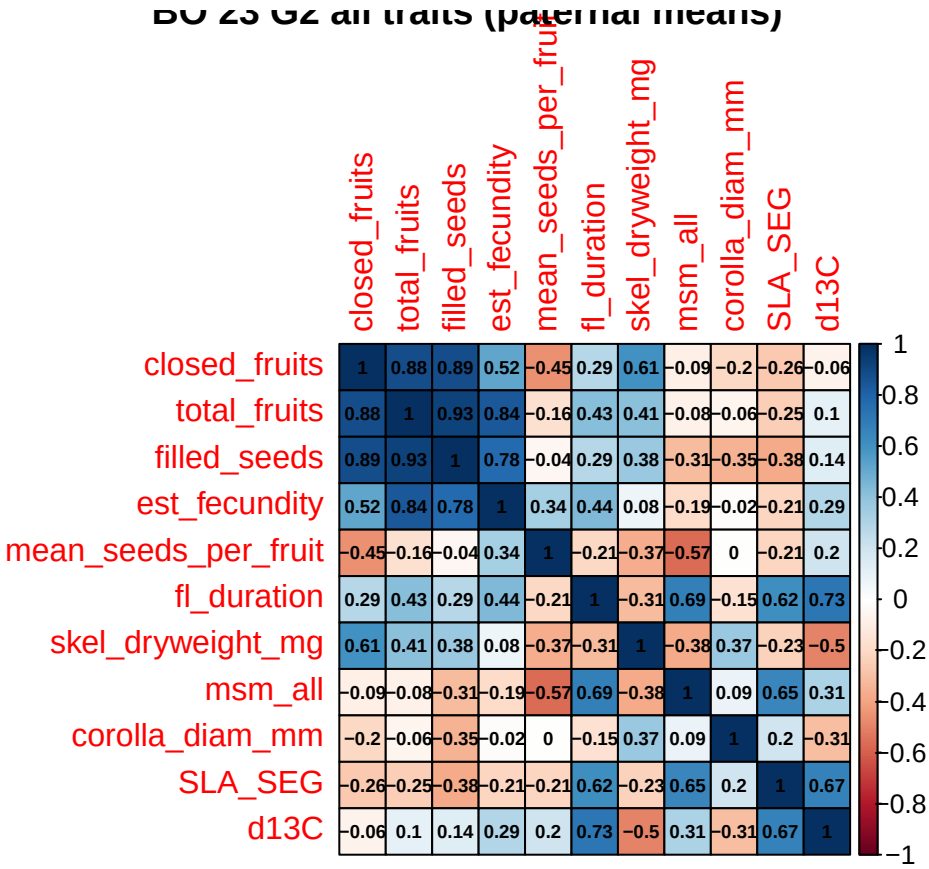

```
##BO 22: Paternal means
corrplot(cor(BO_fit_22 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundity, mean_seeds_p
```




```
##BO 23 G1: Paternal means
corrplot(cor(BO_fit_23_G1 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundity, mean_seed
```




```
##BO 23 G2: Paternal means
corrplot(cor(BO_fit_23_G2 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundity, mean_seed
```



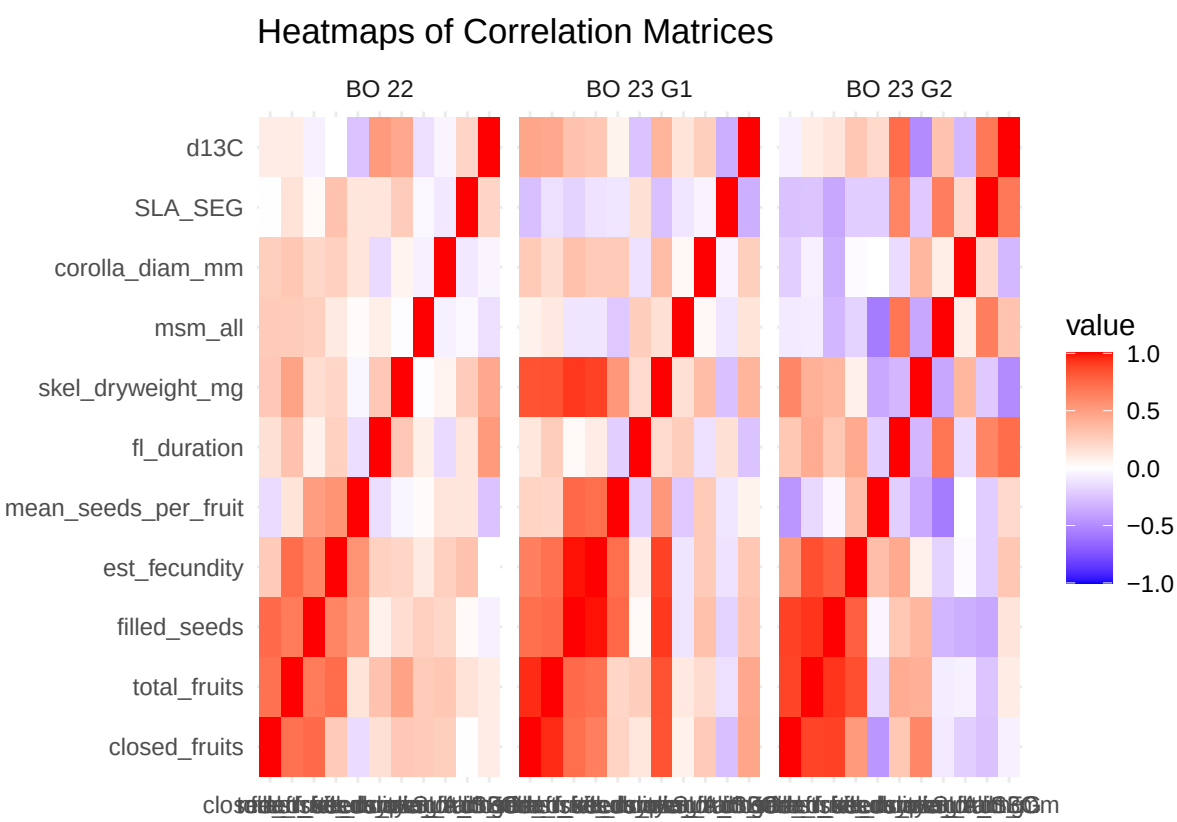
Comparison of matrices

```
# Extract correlation matrices
cor_matrix_22 <- cor(BO_fit_22 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundity, mean
cor_matrix_23_G1 <- cor(BO_fit_23_G1 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundity
cor_matrix_23_G2 <- cor(BO_fit_23_G2 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundity

# Flatten correlation matrices
cor_22_flat <- as.vector(cor_matrix_22)
cor_23_G1_flat <- as.vector(cor_matrix_23_G1)
cor_23_G2_flat <- as.vector(cor_matrix_23_G2)

# Combine into a data frame for comparison
cor_comparison <- data.frame(
  Correlation = c(cor_22_flat, cor_23_G1_flat, cor_23_G2_flat),
  Dataset = rep(c("BO 22", "BO 23 G1", "BO 23 G2"), each = length(cor_22_flat))
)

# Summary statistics
cor_comparison %>%
  group_by(Dataset) %>%
  summarise(
    Mean = mean(Correlation, na.rm = TRUE),
    SD = sd(Correlation, na.rm = TRUE)
  )
```

```
## # A tibble: 3 x 3
```

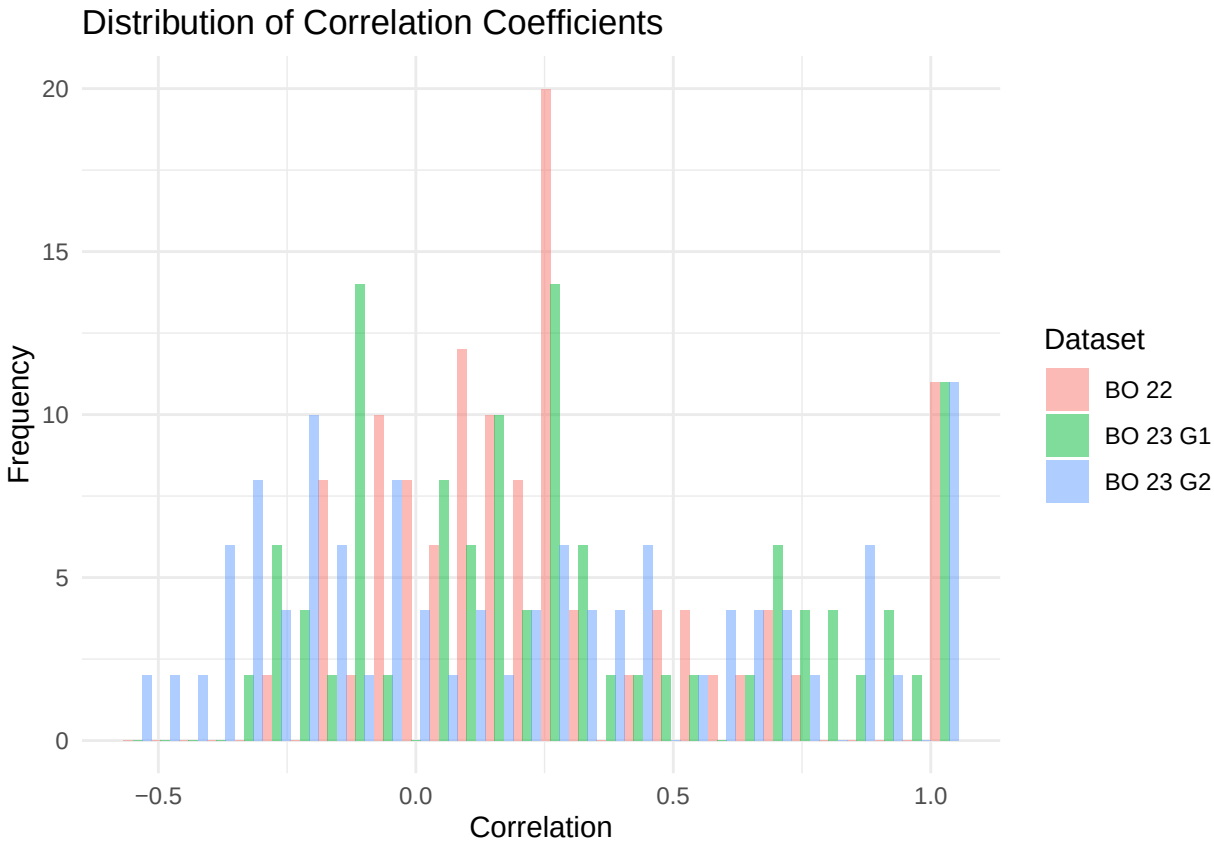
```
##   Dataset   Mean    SD
##   <chr>    <dbl> <dbl>
## 1 BO 22    0.266 0.326
## 2 BO 23 G1 0.315 0.407
## 3 BO 23 G2 0.204 0.468
```

```r
# Melt correlation matrices for ggplot
melt_cor_matrix <- function(cor_matrix, title) {
  melted <- melt(cor_matrix)
  melted$Dataset <- title
  return(melted)
}

cor_22_melt <- melt_cor_matrix(cor_matrix_22, "BO 22")
cor_23_G1_melt <- melt_cor_matrix(cor_matrix_23_G1, "BO 23 G1")
cor_23_G2_melt <- melt_cor_matrix(cor_matrix_23_G2, "BO 23 G2")

combined_melt <- rbind(cor_22_melt, cor_23_G1_melt, cor_23_G2_melt)

# Plot heatmaps
ggplot(combined_melt, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile() +
  facet_wrap(~Dataset) +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white", midpoint = 0, limit = c(-1, 1)) +
  theme_minimal() +
  labs(title = "Heatmaps of Correlation Matrices", x = "", y = "")
```




```r
# Plot histograms of correlations
ggplot(cor_comparison, aes(x = Correlation, fill = Dataset)) +
```

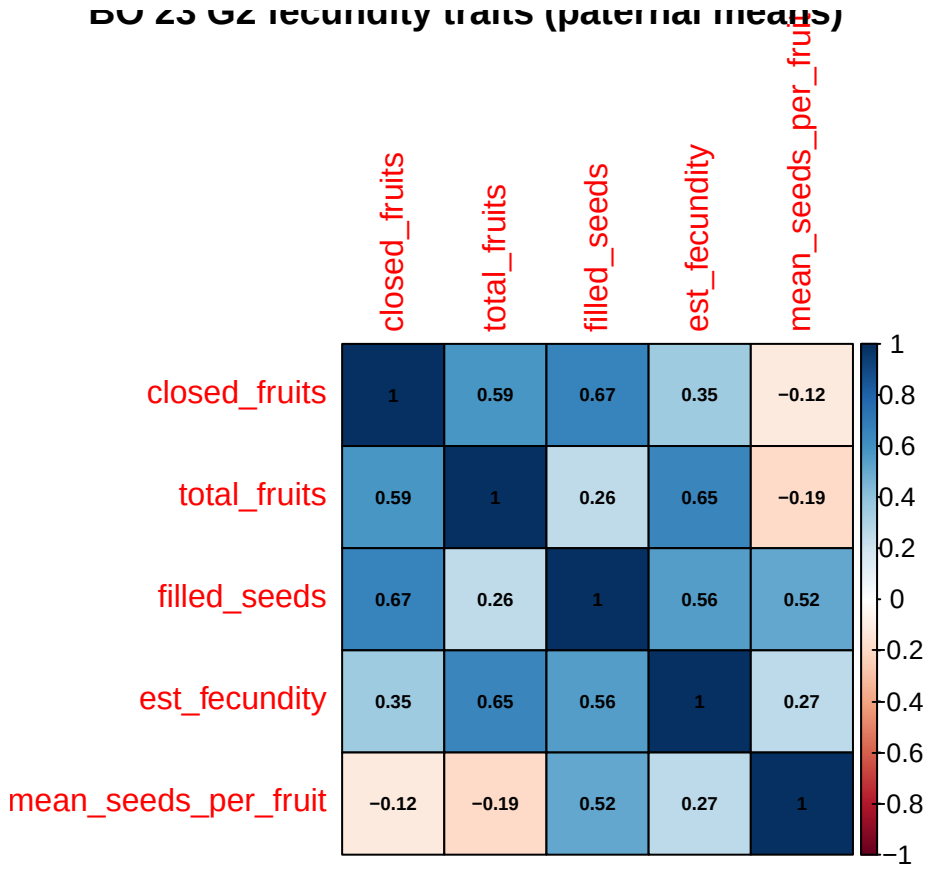
```r
  geom_histogram(position = "dodge", bins = 30, alpha = 0.5) +
  labs(title = "Distribution of Correlation Coefficients", x = "Correlation", y = "Frequency") +
  theme_minimal()
```



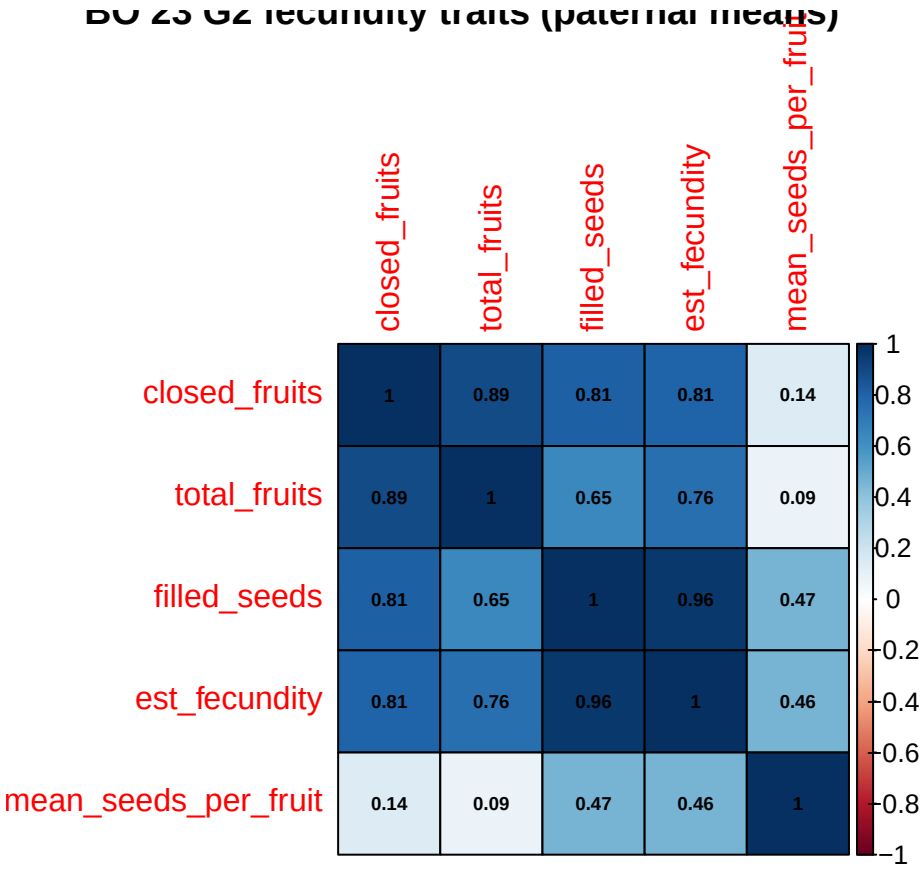

```r
# Aggregate data by donor
mean_donor_traits_22 <- BO_fit_22%>%
  filter(est_fecundity != 0) %>%  # Filter out rows where est_fecundity is zero
  group_by(Donor) %>%             # Group by donor
  summarise(
    closed_fruits = mean(closed_fruits, na.rm = TRUE),
    total_fruits = mean(total_fruits, na.rm = TRUE),
    filled_seeds = mean(filled_seeds, na.rm = TRUE),
    est_fecundity = mean(est_fecundity, na.rm = TRUE),
    mean_seeds_per_fruit = mean(mean_seeds_per_fruit, na.rm = TRUE),
    fl_duration = mean(fl_duration, na.rm = TRUE),
    skel_dryweight_mg = mean(skel_dryweight_mg, na.rm = TRUE),
    est_fecundity = mean(est_fecundity, na.rm = TRUE),
    msm_all = mean(msm_all, na.rm = TRUE),
    corolla_diam_mm = mean(corolla_diam_mm, na.rm = TRUE),
    SLA_SEG = mean(SLA_SEG, na.rm = TRUE),
    d13C = mean(d13C, na.rm = TRUE)
  )

# Aggregate data by donor
mean_donor_traits_23_G1 <- BO_fit_23_G1%>%
  filter(est_fecundity != 0) %>%  # Filter out rows where est_fecundity is zero
  group_by(Donor) %>%             # Group by donor
```

```r
  summarise(
    closed_fruits = mean(closed_fruits, na.rm = TRUE),
    total_fruits = mean(total_fruits, na.rm = TRUE),
    filled_seeds = mean(filled_seeds, na.rm = TRUE),
    est_fecundity = mean(est_fecundity, na.rm = TRUE),
    mean_seeds_per_fruit = mean(mean_seeds_per_fruit, na.rm = TRUE),
    fl_duration = mean(fl_duration, na.rm = TRUE),
    skel_dryweight_mg = mean(skel_dryweight_mg, na.rm = TRUE),
    est_fecundity = mean(est_fecundity, na.rm = TRUE),
    msm_all = mean(msm_all, na.rm = TRUE),
    corolla_diam_mm = mean(corolla_diam_mm, na.rm = TRUE),
    SLA_SEG = mean(SLA_SEG, na.rm = TRUE),
    d13C = mean(d13C, na.rm = TRUE)
  )

# Aggregate data by donor
mean_donor_traits_23_G2 <- BO_fit_23_G2%>%
  filter(est_fecundity != 0) %>%  # Filter out rows where est_fecundity is zero
  group_by(Donor) %>%             # Group by donor
  summarise(
    closed_fruits = mean(closed_fruits, na.rm = TRUE),
    total_fruits = mean(total_fruits, na.rm = TRUE),
    filled_seeds = mean(filled_seeds, na.rm = TRUE),
    est_fecundity = mean(est_fecundity, na.rm = TRUE),
    mean_seeds_per_fruit = mean(mean_seeds_per_fruit, na.rm = TRUE),
    fl_duration = mean(fl_duration, na.rm = TRUE),
    skel_dryweight_mg = mean(skel_dryweight_mg, na.rm = TRUE),
    est_fecundity = mean(est_fecundity, na.rm = TRUE),
    msm_all = mean(msm_all, na.rm = TRUE),
    corolla_diam_mm = mean(corolla_diam_mm, na.rm = TRUE),
    SLA_SEG = mean(SLA_SEG, na.rm = TRUE),
    d13C = mean(d13C, na.rm = TRUE)
  )
```

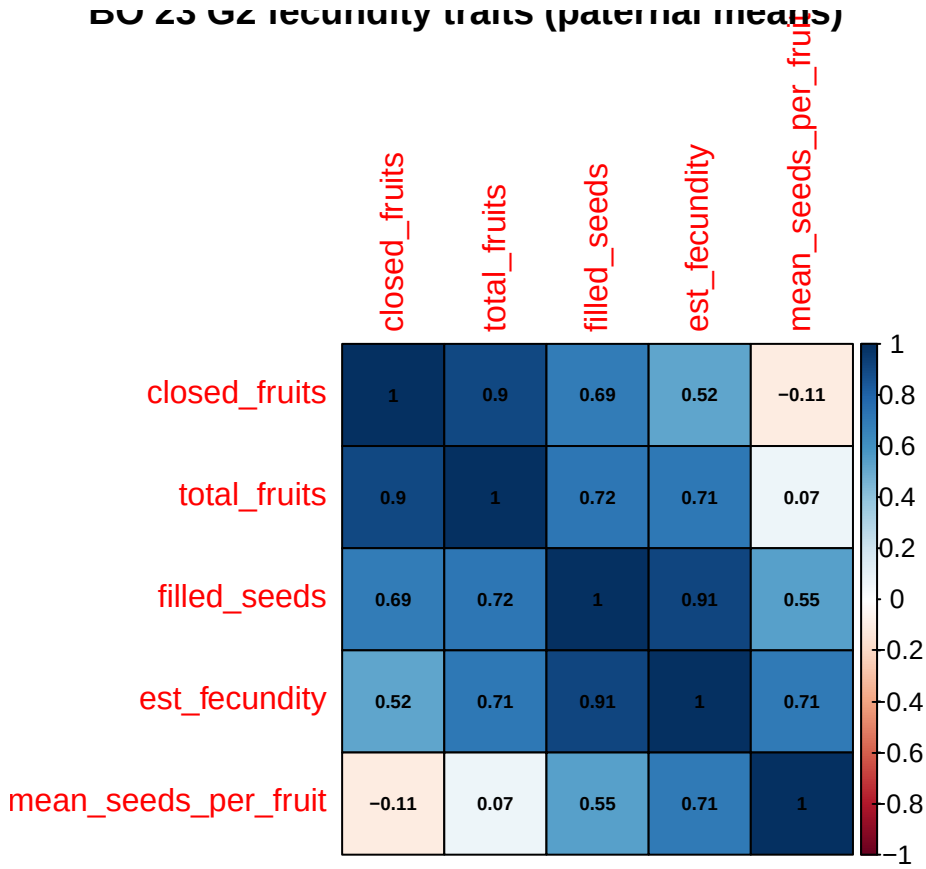
```r
##BO 22: Paternal averages of fitness estimates
corrplot(cor(mean_donor_traits_22 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundity, m
```




```
##BO 23 G1: Paternal averages of fitness estimates
corrplot(cor(mean_donor_traits_23_G1 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundit
```

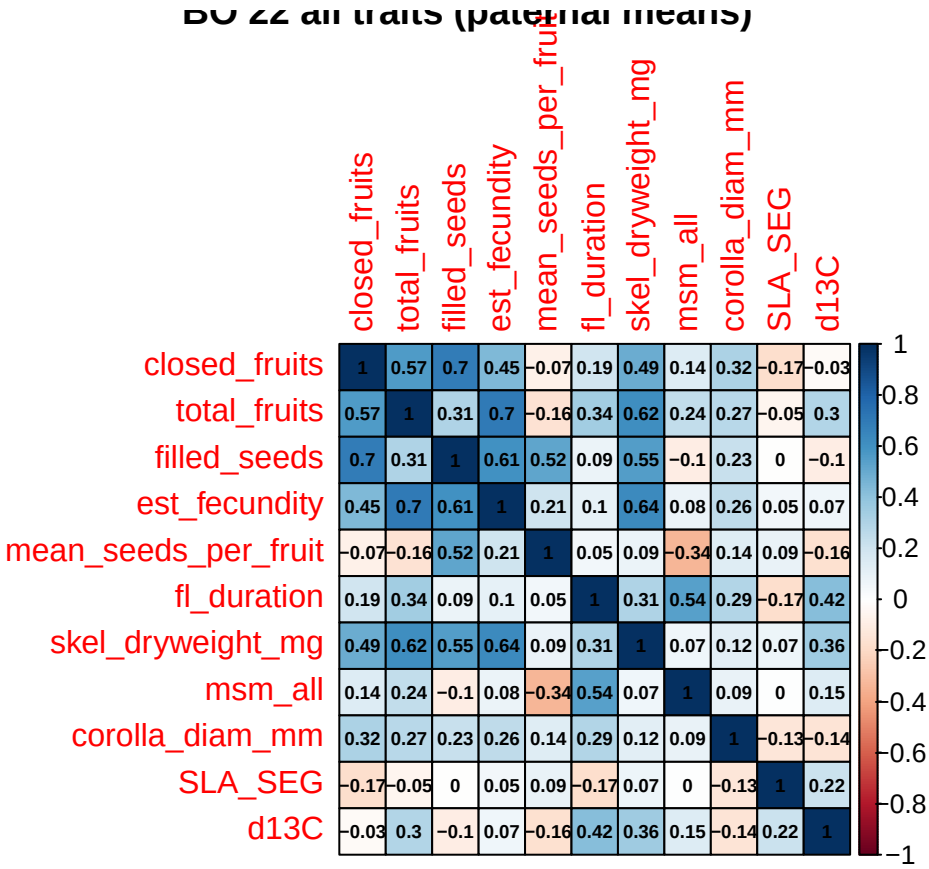



```
##BO 23 G2: Paternal averages of fitness estimates
corrplot(cor(mean_donor_traits_23_G2 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundit
```

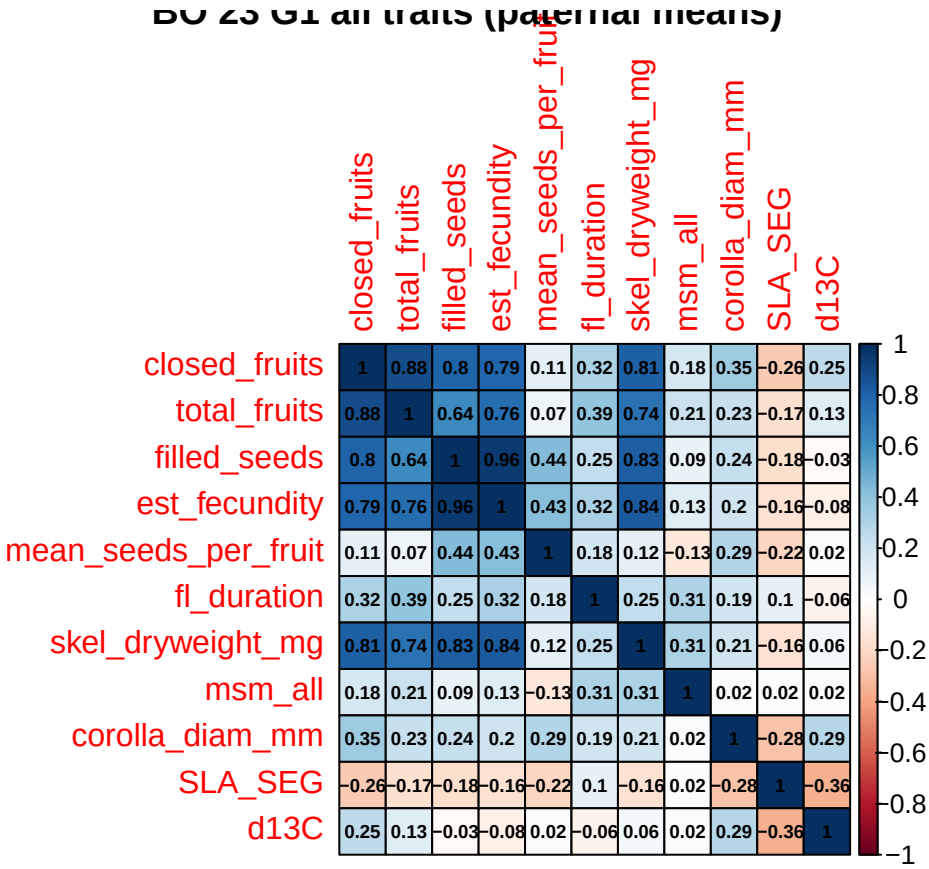



```
#as the number of closed fruits increases, the mean number of seeds per fruit tends to decrease, as tot
##This correlation is only apparent when looking at the paternal means
```

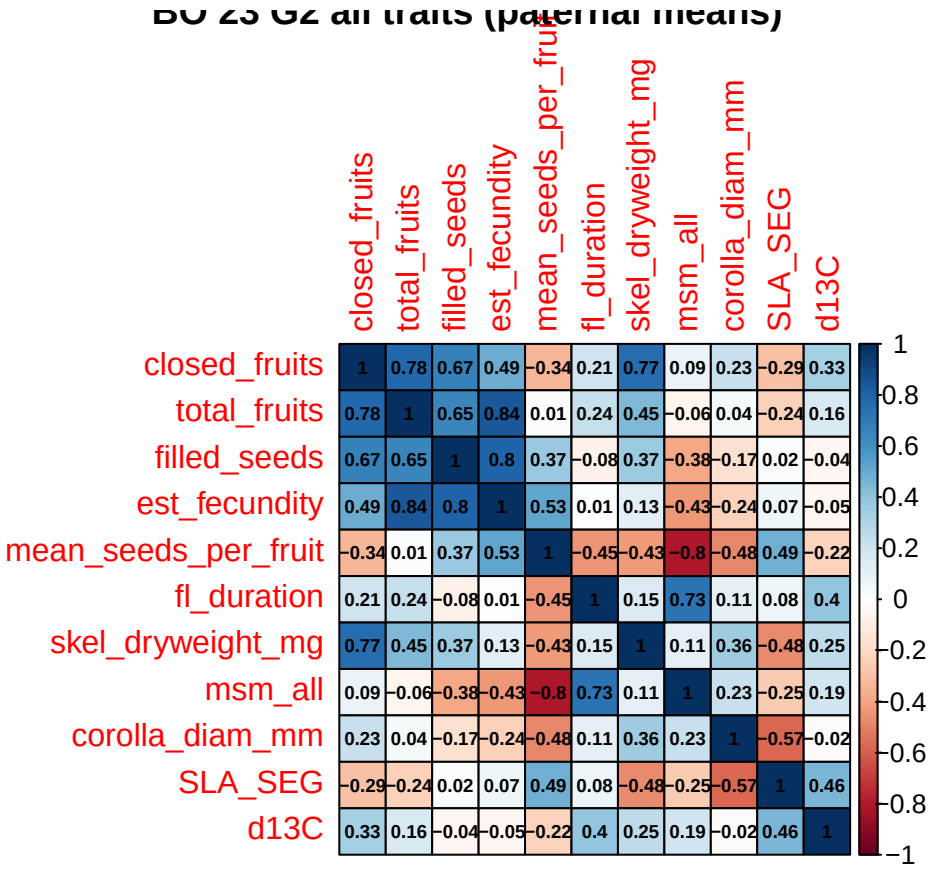
```
##BO 22: Paternal averages of fitness estimates
corrplot(cor(mean_donor_traits_22 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundity, m
```




```
##BO 23 G1: Paternal averages of fitness estimates
corrplot(cor(mean_donor_traits_23_G1 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundit
```




```
##BO 23 G2: Paternal averages of fitness estimates
corrplot(cor(mean_donor_traits_23_G2 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecundit
```



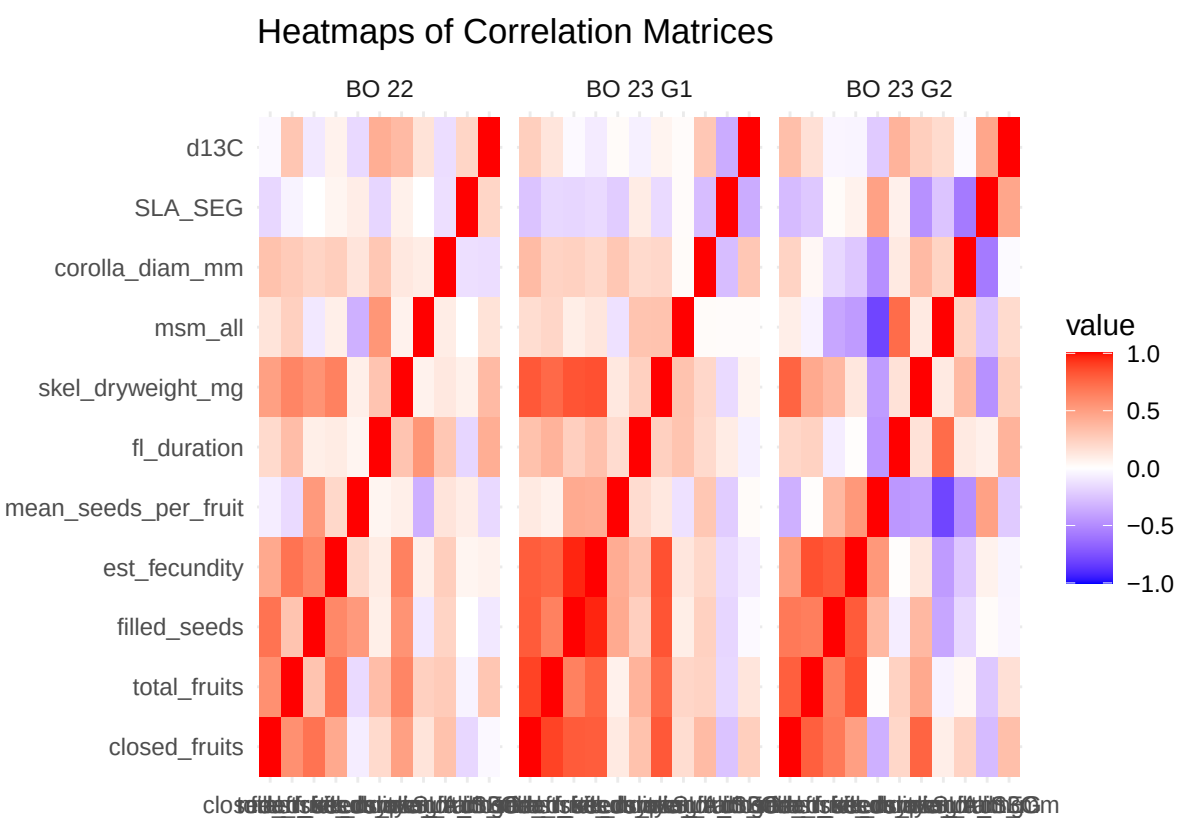
Comparison of matrices

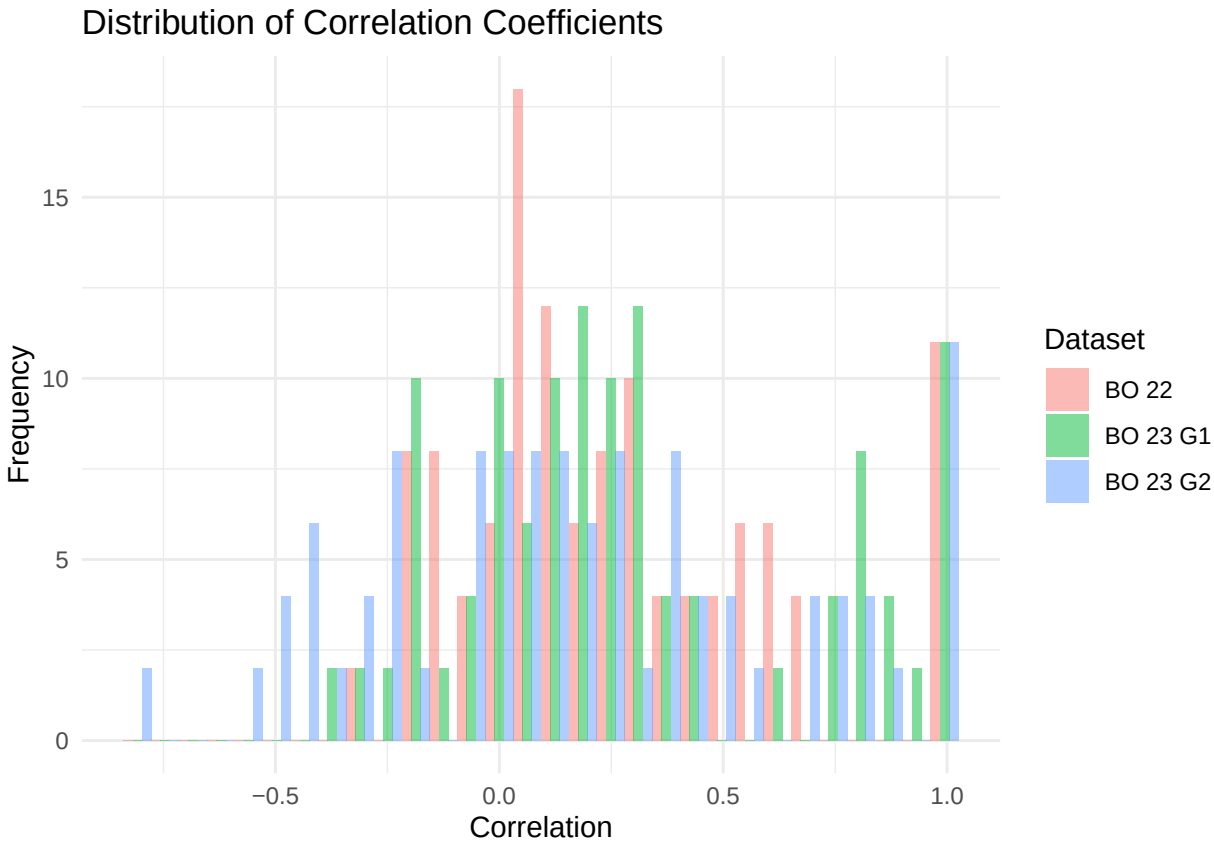
```
# Extract correlation matrices
cor_matrix_22 <- cor(mean_donor_traits_22 %>% select(closed_fruits, total_fruits, filled_seeds, est_fecu
cor_matrix_23_G1 <- cor(mean_donor_traits_23_G1 %>% select(closed_fruits, total_fruits, filled_seeds, es
cor_matrix_23_G2 <- cor(mean_donor_traits_23_G2 %>% select(closed_fruits, total_fruits, filled_seeds, es

# Flatten correlation matrices
cor_22_flat <- as.vector(cor_matrix_22)
cor_23_G1_flat <- as.vector(cor_matrix_23_G1)
cor_23_G2_flat <- as.vector(cor_matrix_23_G2)

# Combine into a data frame for comparison
cor_comparison <- data.frame(
  Correlation = c(cor_22_flat, cor_23_G1_flat, cor_23_G2_flat),
  Dataset = rep(c("BO 22", "BO 23 G1", "BO 23 G2"), each = length(cor_22_flat))
)

# Summary statistics
cor_comparison %>%
  group_by(Dataset) %>%
  summarise(
    Mean = mean(Correlation, na.rm = TRUE),
    SD = sd(Correlation, na.rm = TRUE)
  )
```

```
## # A tibble: 3 x 3
```

```
##   Dataset   Mean    SD
##   <chr>    <dbl> <dbl>
## 1 BO 22    0.263 0.336
## 2 BO 23 G1 0.302 0.384
## 3 BO 23 G2 0.187 0.451
```

```r
# Melt correlation matrices for ggplot
melt_cor_matrix <- function(cor_matrix, title) {
  melted <- melt(cor_matrix)
  melted$Dataset <- title
  return(melted)
}

cor_22_melt <- melt_cor_matrix(cor_matrix_22, "BO 22")
cor_23_G1_melt <- melt_cor_matrix(cor_matrix_23_G1, "BO 23 G1")
cor_23_G2_melt <- melt_cor_matrix(cor_matrix_23_G2, "BO 23 G2")

combined_melt <- rbind(cor_22_melt, cor_23_G1_melt, cor_23_G2_melt)

# Plot heatmaps
ggplot(combined_melt, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile() +
  facet_wrap(~Dataset) +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white", midpoint = 0, limit = c(-1, 1)) +
  theme_minimal() +
  labs(title = "Heatmaps of Correlation Matrices", x = "", y = "")
```




```r
# Plot histograms of correlations
ggplot(cor_comparison, aes(x = Correlation, fill = Dataset)) +
```

```r
geom_histogram(position = "dodge", bins = 30, alpha = 0.5) +
labs(title = "Distribution of Correlation Coefficients", x = "Correlation", y = "Frequency") +
theme_minimal()
```